KUSH SHARMA

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PROFILE

MSc Artificial Intelligence student at the University of the West of England (UWE), with a strong engineering background and hands-on experience in machine learning, data analysis, and simulation-driven optimisation. Former Design Engineer at Intute AI, experienced in applying data-driven and analytical techniques to real-world engineering problems.

EDUCATION

MSc Artificial Intelligence, University of the West of England (UWE), Bristol, UK (2025–2026)

Relevant modules: Machine Learning Algorithms, AI for Search & Optimisation, Statistical Inference, Cloud Computing

B.E. Electrical Engineering, Thapar Institute of Engineering and Technology, Patiala, India (2021–2025)

TECHNICAL SKILLS

Programming & Data: Python, SQL, C/C++, Pandas, NumPy

Machine Learning: Supervised & Unsupervised Learning, Feature Engineering, Model Evaluation, Cross-Validation, Hyperparameter Tuning

AI & Optimisation: Search Algorithms, Heuristics, Optimisation Techniques

Visualisation: Matplotlib, Seaborn

Simulation & Engineering Tools: Ansys MotorCAD (EMag, Thermal, Lab), MATLAB

PROJECTS

Credit Risk Prediction (Python): github.com/Kushs22/home-credit-default-risk-ml

- Built an end-to-end ML pipeline on the Home Credit Default Risk dataset (307k+ records, 122 features) including preprocessing, missing-value handling, encoding, and feature engineering (expanded to 230 features).
- Trained and evaluated Logistic Regression and Decision Tree models, achieving ROC-AUC ≈ 0.75 and 68.9% accuracy while improving recall for detecting high-risk borrowers in an imbalanced classification setting.

Travelling Salesman Problem - Search & Optimisation: github.com/Kushs22/tsp-search-optimisation-AI

- Implemented exact and heuristic optimisation algorithms to solve the Travelling Salesman Problem for a 5-city network, evaluating 120 routes via brute-force search to determine the optimal path.
- Developed a distance-matrix optimisation model and greedy heuristic approach, analysing runtime vs solution quality trade-offs for combinatorial optimisation problems.

Wireless EV Charging using Solar Power (Undergraduate Project)

- Designed and prototyped a solar-powered wireless EV charging setup integrating power electronics, control, and inductive coils.
- Developed and tested a working prototype demonstrating efficient wireless energy transfer for sustainable EV charging applications.

EXPERIENCE

Design Engineer, Intute AI Technologies Pvt. Ltd. (Jan 2025–May 2025

- Conducted electromagnetic and thermal FEA simulations of BLDC, IPM, and PMSM motors using ANSYS MotorCAD for electric vehicle traction systems.
- Optimised motor design parameters (slot geometry, magnet configuration, phase advance), improving efficiency from ~74% to ~90.4%.
- Performed thermal analysis with water-jacket cooling, maintaining safe operating temperatures (≈33–35°C) under load conditions.
- Analysed copper, iron, and windage losses to support data-driven motor design decisions and prepared technical reports for engineering teams.